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Cited Text Span Extraction for Scientific Question Answering

Bachelor's Thesis

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1. Introduction

Large Language Models (LLMs) have fundamentally transformed how we look up information and obtain answers to questions, becoming an increasingly important tool, not only in everyday life, but also across a range of professional fields. With the usage of such systems, the need to verify the factual correctness and easily do further research on the underlying sources arises, especially when answers lack important details or provide a depth of explanation that is misaligned with the user's existing knowledge. This is especially important in high-stakes fields such as medicine, law or scientific research, where seemingly plausible but unbacked or false claims could lead users to draw false conclusions.

One increasingly popular method is the approach of Retrieval Augmented Generation (RAG), replacing zero-shot LLM solutions with a two-step process. Firstly, relevant sources in the form of documents or other multi-media data are retrieved and then fed into the generating model for it to use as ground truth for answer generation and later to enable citation extraction. Retrieving correct and precise text-spans alongside the generated answer is not a trivial task, and with domain-specific requirements widely varying, such systems are part of active research [1] [2]. One system applying such a RAG approach is SQuAI [3], aimed at answering complex open-domain scientific questions on its knowledgebase containing 2.3 million full-text scientific papers. For further performance improvements, it introduces a multi-agent pipeline, enhancing document retrieval capabilities as well as providing document-level backlinks for parts of its generated answer. This system, while providing some backlinking, does still fall short of providing more granular, sentence-level cited text spans, which have proven to be much more useful to users, as they are quicker and easier to verify, enhancing trustworthiness and usability of the system.

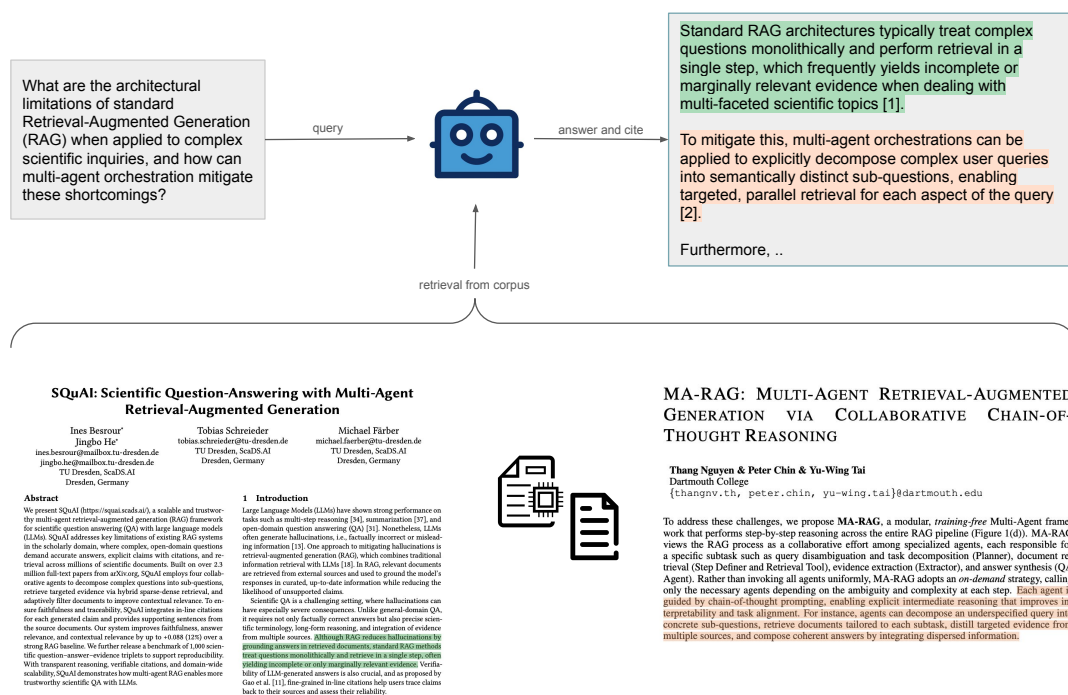


Figure 1.1.: Sentence level cited text span attribution

Therefore, this thesis will focus on benchmarking different cited text span extraction methods to enhance source attribution, leading to the following core contributions:

1. Curate a dataset of high-quality, multi-hop scientific questions building upon a large-scale retrieval corpus that require complex reasoning capabilities in order to be answered and closely resemble real-world scientific inquiry needs.
2. Introduce a hierarchy of different cited text span extraction methods ranging from simple lexical and semantic matching up to complex reasoning capabilities, as well as strategic combinations of such, to allow for tradeoffs in accuracy and efficiency.
3. Apply an automated framework to benchmark the retrieval and answering capabilities of non-parametric, retrieval- and evidence-based text generation systems on the proposed dataset as well as evaluate the extraction methods in terms of efficiency, correctness and quality of source attribution using NLI models and LLM-as-a-judge approaches.

2. Related Work

This chapter provides a comprehensive review of prior work across three key dimensions relevant for the development of a scientific QA system capable of providing cited text spans as evidence for its generated answer. First, it examines existing scientific QA datasets with respect to their suitability for SQuAI. It then compares different established cited text span extraction paradigms and their tradeoffs in terms of precision, efficiency and computational requirements before discussing automatic approaches for evaluating text span attribution quality in long-form, multi-hop settings.

2.1. Datasets

In order to improve upon the existing SQuAI system, a highly specialized dataset of open-domain, long form scientific questions is needed. Due to the scarcity and high cost of curating such datasets, only a limited number of suitable options are available. With SQuAI being designed as a system to answer complex scientific inquiries based on the unarXive 2022 Dataset [4], further requirements are introduced, namely:

1. **Domain Alignment:** Questions must fall under the knowledge scope of the underlying knowledgebase, in order for the system to be able to provide cited text spans as proof alongside its answer. In terms of the SQuAI System, this boils down to questions targeting physics, mathematics and computer science that are also part of the papers in the unarXive 2022 Dataset.
2. **High Complexity:** Questions must go beyond simple factoid questions and require complex reasoning capabilities to answer, as questions posed by scientists usually require synthesizing, interpreting and evaluating data, requiring high cognitive abilities to interact with. In order to be able to test if the SQuAI system can be used under such circumstances, the questions in the dataset must also have a similar degree of complexity.
3. **Multi-Hop:** Questions must require the information in multiple documents to be answered, in order to test the ability of the system to synthesize information from multiple sources, as is required in naturally occurring scientific research.
4. **Decontextualization:** Questions must be answerable in an open-domain setting and cannot contain any kind of reference to specific papers and their contents that lose meaning or become ambiguous outside of the context of their paper. As a result, questions must contain enough context such that they are standalone answerable.
5. **Text format:** Questions can only target information in a paper that is present in the form of text, as current versions of the SQuAI system do solely rely on textual information, not taking into account images, graphs or similar.
6. **Ground Truth:** In order to evaluate the correctness of the model's answer, a dataset must provide a ground answer alongside each question.

However, comparing some of the most prominent datasets shows that none fits all discussed criteria, finally resulting in the need for a newly curated dataset.

Dataset	Domain Alignment	High Complexity	Multi-Hop	Decontextualization	Text format	Ground Truth	Sufficient Dataset size (>500)
Qasper [5]	partial	shallow	no	no	yes	yes	yes
ALCE [6]	no	shallow	yes	yes	yes	yes	yes
SPIQA [7]	yes	complex	no	partially	no	yes	yes
QASA [8]	yes	complex	no	no	yes	yes	yes
ExpertQA [9]	yes	very complex	yes	yes	yes	yes	no ¹
SciDQA [10]	yes	very complex	no	no	no	yes	yes
PeerQA [11]	yes	very complex	no	no	yes	yes.	yes

Table 2.1.: Comparison of Scientific QA Datasets against SQuAI Requirements

Abbreviations: CS = Computer Science; ML = Machine Learning; NLP = Natural Language Processing.

2.2. Cited Text Span Extraction

Context extraction methods can be most broadly divided into pre- and post-hoc extraction methods, depending on the time the context attribution happens. While pre-hoc systems such as QASPER [5] and QASA [8] extract citable passages before and during runtime of the question-answering system, post-hoc systems search and attach supporting evidence afterwards. Post-hoc systems have been shown to outperform pre-hoc systems, achieving higher citation coverage and answer correctness as well as lowering model hallucination [12]. While a number of systems rely on zero-shot LLM solutions for context extraction [13] [14] [15], work on the DECOMPTUNE [16] system has shown that decomposing facts and decontextualizing answer sentences before attributing source text spans with LLM prompting can yield significant improvements, especially when applied to answers for long-form, multi-hop questions. In order to eliminate the probability of hallucinated model output, models can be finetuned [17] or the answer space restricted to the start and end token of the cited text span, forcing strict citation [18].

So-called white-box methods rely on access to the model’s internal weights, rendering them incompatible with proprietary LLMs. The LAQuer [19] system, as well as earlier token-embedding approaches it is built upon [20], analyzes internal hidden states to extract the most likely text span that a user-highlighted part of the LLM’s answer is based upon. To pinpoint a certain text span, the system calculates the maximum cosine similarity between individual tokens in the source text and the averaged hidden representation of the highlighted span, for which attribution is sought. The system then extracts the precise quote from the surrounding window of text. However, this method has been shown by the authors to perform significantly worse than LLM Prompting, with drastically lower accuracy and missing suitable citations, especially in larger search spaces, e.g. longer text. Similarly, AttnAttribute [21] identifies "Attention Heads", tracking the model’s internal attention values during generation time. Just like LAQuer, it then extracts text spans from the

¹question-answer pairs are broadly distributed across diverse fields of study, resulting in insufficient topical overlap with unarXive 2022

passage surrounding the anchor with the highest model attention. Benchmarks made as part of the development of DECOMPTUNE [16] concluded that both of those methods performed significantly worse than LLM Prompting.

Recent advancements have explored generative approaches. Rather than retrieving and reranking text spans, such systems generate them. In order to prevent hallucinations, the output vocabulary of the systems is typically restricted to the underlying retrieval corpus by applying constrained decoding mechanisms, such as prefix trees or FM-indices [22]. GERE [23], as one such system using a prefix tree, applies this concept of restricted output in the document retrieval step, generating document titles for selecting appropriate source texts, as well as sentence identifiers mapped to individual sentences. This system, while removing the need for a massive document index as well as showing significant improvements in precise and correct text span extraction compared to baselines such as Dense Passage Retrieval (DPR) and BM25, requires the prefix tree to be precomputed and stored in memory, making it highly inflexible for changes in the underlying corpus. Similarly, MRR-FV [24] further improves this approach with multi-hop reasoning capabilities. Instead of generating sentence identifiers, it generates sentences directly, leveraging the natural language understanding capabilities of LLMs, leading to improvements in precision and quality of attributed spans, but also further bloating the prefix tree, making the system even harder to scale.

Despite the diversity of proposed paradigms, current work mostly evaluates them in isolation, restricting comparison to baselines or direct predecessors on heterogeneous tasks, making broad comparisons for practical applicability unfeasible. Consequently, this work introduces a hierarchy of extraction methods, increasing in both reasoning capabilities and computational requirements, benchmarking both accuracy and efficiency on a unified framework, allowing for systematic trade-off analysis between paradigms in the context of open-domain multi-hop scientific QA.

2.3. Span Evaluation

While human annotation has always been the gold standard for assessing natural language generation (NLG) tasks, relying solely on it is a bottleneck for developing models of increasing size and complexity. While providing feedback with extensive reasoning and natural language understanding, human evaluation suffers from being costly, time-intensive and, depending on the specific task, highly subjective. Thus, lowering or even replacing the need for human evaluation completely is an important problem.

Early metrics heavily rely on measuring the lexical overlap between text spans, for instance of a human-written ground truth and an LLM-generated answer for the same questions. The most prominent examples include BLEU [25] and ROUGE [26], both being n-gram based metrics. While the former is precision-oriented, measuring how many of its n-grams also appear in the human reference span, the latter applies a recall-oriented approach, measuring how many n-grams of the human reference are also present in the LLM-generated text span. Being computationally inexpensive and language independent, these metrics have established themselves as rigid industry standards and baselines for other methods. Their alignment with human judgement, however, has been shown to

be low, suffering from the lack of semantic understanding, leading to penalizing valid paraphrasing as well as missing contradictory or hallucinated content.

Partly overcoming those limitations, embedding-based metrics like BERTScore [27] and MoverScore [28] emerged, measuring semantic instead of lexical similarity. BERTScore does one-to-one greedy matching on vector embeddings, providing precision, recall and F1 score metrics, just like BLEU and ROUGE, but on an embedding level. MoverScore improves upon this idea, utilizing Earth Mover’s Distance (EMD) to allow for introducing one-to-many relationships, allowing for matching of single words with phrases, so-called soft matching, as well as introducing Inverse Document Frequency (IDF) for enhanced lexical matching, to better reflect the importance of specific keywords to be included. Though significantly more expensive to compute, those metrics show significant improvements in semantic robustness as well as intrinsic context awareness, due to their vector embeddings. However, both of those metrics still require a reference answer to compare to. BartScore [29] eliminates this requirement, measuring the conditional probability that one text span is generated from another using an encoder-decoder model, arguing that the probability is higher if the claims in the answer are also deeply rooted in the source text.

Besides similarity-based metrics, other research suggests approaching the task from a logical entailment perspective. The AIS [30] framework proposed a unified conceptual definition for such, arguing that a statement is considered Attributable to Identified Sources (AIS) iff completely corroborated by a source. In order to verify this, the authors imposed additional constraints being 1) that the claim made in an answer sentence must be standalone interpretable, thus decontextualized and 2) the information in the text span must support all the information present in the answer span it is attributed to. Previously discussed similarity-based metrics like BLEU, ROUGE or token-level F1, as well as solely relying on an out-of-the-box Natural Language Inference (NLI) model, have been shown to have low correlation with human judgement for both factual consistency and logical entailment assessment [31] [32]. To address these limitations, systems relying on NLI models like FactCC [32] argued that evaluating logical entailment on sentence-to-sentence level is insufficient, since information in sources might be spread across multiple passages. As a result, these systems increased their evaluation scope to the whole source text, leading to better judgement capabilities, but also increase in model input. The authors of FactCC observed that a majority of misclassifications were commonsense mistakes; the TRUE benchmark [31] observed improvements with increasing model size but also degradation with longer input, which is further supported by research on the SummaC [33] framework. To shrink input size, SummaC increased evaluation granularity by dividing both the source text and the model answer into smaller units, checking each combination of them for entailment and combining the obtained scores.

The Q² system [34] proposed using a Question-Generation - Question-Answering (QG-QA) method, generating questions to which the LLM-generated answer is the solution. Those questions are then answered with the original source document and lastly the so obtained answers are then compared with the original LLM answer using an NLI model. The authors presented significant improvements compared to baseline methods, further enhanced by the TRUE benchmark by applying an ensemble approach utilizing ANLI (standard NLI), SummaC (increased granularity NLI) as well as the Q² (QG-QA) method. The ensemble was shown to beat every standalone evaluation metric previously

discussed on all datasets benched in TRUE, emphasizing the need for combined, complementary evaluation metrics.

Recently, the evaluation paradigm has shifted to LLM-as-a-judge approaches. While early versions like GPTScore [35] tried evaluating based on model internals, using the conditional token probability, thus the confidence of a model in a given answer, they have quickly been shown to be much less effective than prompting the LLM directly to grade [36]. The performance of LLMs in evaluation tasks can additionally be enhanced by applying a Chain-of-thought approach, prompting the model to break down complex tasks into simpler ones and adding intermediate reasoning steps. Closely resembling human cognitive processes, this method yields significant improvements, especially with larger models and problems of increasing complexity [37], reaching agreement rates comparable to human judges for open-ended multi-turn dialogues [38], thus eliminating human annotation constraints.

When applied to long-form text containing multiple pieces of information, binary judgements become insufficient as they cannot reflect partial attribution. FactScore [39] addresses this by breaking down answers into atomic facts using an LLM, obtaining a percentage measure of supportedness. AttrScore [40] introduces additional evaluation criteria, such as measures for attribution, extrapolation and contradiction, in order to further characterize the type of attribution error. Similarly, CiteEval [41] introduces penalties for attributions containing unnecessary noise and redundant citations while advocating for evaluation against full source documents, in order to not lose information in the surrounding context. The RAGAS [42] framework operationalized these findings, providing a suite of automated metrics like faithfulness, answer relevance and context relevance, without requiring human annotation. However, the importance of penalizing undesired behaviors is further emphasized by the MT-Benchmark [38], showing that LLMs have a multitude of biases, such as favoring longer and LLM-generated text spans or switching preferences between answers when presented in a different order. Furthermore, LLMs perform remarkably poorly with fine-grained information, with a majority of mistakes being caused by overlooking numbers and entities or hallucinating logical connections [43]. Using increasingly large models additionally introduces high computational cost requirements. To address this, systems like ARES [44] rely on LLM-generated, domain-specific, synthetic training data to finetune smaller models. Using Prediction-Powered-Inference (PPI), ARES leverages a small human-annotated dataset to calculate the LLM’s annotation error on the synthetic data, the "rectifier", to provide statistical confidence bounds and enhance its evaluation, beating existing zero-shot solutions like RAGAS.

Automatic evaluation of cited text span attribution remains a challenging task. Despite the wide range of approaches, no single metric being capable of reliably and comprehensively assessing attribution performance as a whole has emerged. Increasingly complex tasks, such as multi-hop questions and challenging application domains like scientific QA, introduce further evaluation dimensions. Consequently, combining multiple evaluation metrics with complementary strengths becomes necessary to measure task-specific aspects within a unified framework.

3. Methodology and Approach

3.1. Dataset

To curate a dataset of high-quality, expert-level scientific question-answer pairs, a fully automated pipeline is applied. It consists of three consecutive steps: document selection, question answer generation and quality insurance, in order to ensure that generated questions and answers are scientifically rigorous and strictly dependent on multi-document information synthesis to be answered, as well as reliably providing associated ground truth documents alongside.

3.1.1. Document selection

To match the underlying document corpus, selectable papers are restricted to those of the unarXive 2022 dataset, yet for other application this could be expanded at will. A randomly selected anchor paper as well as its cited references serve as the foundation for generating a single question-answer pair. For each paper, a document-level SPECTER [45] vector embedding is retrieved via the Semantic Scholar academic graph API [46]. Leveraging citation graphs, these vectors provide dense representations of a paper’s title and abstract, which are subsequently used to calculate the cosine similarity and thus the semantic similarity of each paper with the anchor document. To ensure topical alignment while excluding highly redundant work, a subset of candidate papers with a cosine similarity between 0.7 and 0.9 is selected. Four of these papers are then chosen at random to be used alongside the anchor paper for generation. Restricting the model input to five papers in total provides a multitude of topical overlaps for the model to choose from without resulting in exceedingly long prompts, mitigating the risk of "Lost in the Middle" effects. If a chosen anchor paper fails to provide at least four appropriate references, it is considered insufficient to be used in the dataset.

3.1.2. Question-Answer generation

While modern LLMs offer increasingly large context windows, usually even above 100,000 tokens, empirical studies have shown severe performance degradation in tasks like complex reasoning, data retrieval and logic problems, accompanied by drastically increased hallucination [47]. In the generation step, the pipeline applies a moonshotai/Kimi-K2.6 [48] model, supporting a context window of approximately 256,000 tokens. In practice however, the "effective context window", in which models only suffer from minimal performance degradations with increasingly long inputs, has been shown to drastically lower by empirical studies [49] and practitioner reports, with some studies estimating it as low as 40% of the maximum supported context window [50]. To ensure high reasoning capabilities, the pipeline applies an adaptive character limit for each paper text, enforcing a total upper bound of 300,000 characters for all papers which conservatively approximated translates to roughly 100,000 tokens² or 40% usage of Kimi-K2.6’s context window. Papers are assigned input

²in standard heuristics for scientific english one token translates to roughly three to four characters

space according to the following formula, allowing for full and efficient utilization of the context window, even if papers are highly variable in length:

$$C_{adaptive} = \left\lfloor C_{base} + \left(\frac{l_{paper}}{\sum_{p \in P_{above}} len(p)} \right) \times \sum_{s \in P_{below}} (C_{base} - len(s)) \right\rfloor \quad (3.1)$$

Paper texts longer than their adaptive character limit are then head and tail truncated, retaining the first and last $\lfloor \frac{C_{adaptive}}{2} \rfloor$ characters, which are the most information dense for scientific papers.

Subsequently the LLM is prompted using a two-step strategy. Firstly the model is tasked with selecting the best pair of papers (A,B), where paper A provides an intermediate entity, method, dataset or similar and paper B evaluating, extending, contradicting or relying on A’s intermediate element. Following that, the LLM is tasked with generating a single Question-Answer pair based on its selected pair, whose answer can only be derived by following a sequential reasoning process across both papers, applying paper A’s intermediate element to paper B. Additionally, the model is instructed to answer the question as precisely as possible, providing a precision-biased subset of the underlying papers used during generation alongside. The complete prompt instructions are provided in Listing A.1.

3.1.3. Quality insurance

In order to ensure the dataset’s integrity two separate evaluation steps are applied, both utilizing a openai/gpt-oss-120b [51] model. The first judge evaluates each Question-Answer pair along six distinct dimension, providing judgement on how good each criteria is satisfied on a three-point ordinal scale.

1. Multi-Hop validity: Does answering the question require combining both papers?
2. Dependency Strength: Measure if the second reasoning step depends on the result of the first one
3. Non-decomposability: Is the question NOT decomposable into independent sub-questions?
4. Evidence distribution: Are both papers required and non-redundant?
5. Answerability: Is the answer fully supported by the provided evidence?
6. Decontextualization: Is the question self-contained and unambiguous?

The complete prompt instructions are provided in Listing A.2.

An analysis of disconnected reasoning by Trivedi et al. [52] demonstrated, that many alleged multi-hop question-answering datasets do in fact not require true multi-hop reasoning, but can instead be solved through reasoning shortcuts, partial evidence or decomposition into independent sub-questions. Using a methodology inspired by the DIRE [52] framework, the second judge verifies, that questions cannot be answered under incomplete evidence, namely a subset $P' \subset P$ of ground

truth papers P , but instead require all ground truth documents for a complete reasoning chain and answer. Thus the test outcome satisfies:

$$(\forall P' \subset P : \neg \text{Pass}(P')) \wedge \text{Pass}(P)$$

The complete prompt instructions are provided in Listing A.3.

The final dataset is then obtained by filtering question-answer pairs to only those, successfully passing both judges.

3.2. Cited text span extraction

As previously discusses, cited text span attribution can be approached through a number of different paradigms, with methods being introduced and applied under vastly different circumstances. Following, a hierarchy of increasingly complex post-hoc extraction methods that will be applied on a unified dataset is introduced. All methods except the native SQuAI span extraction and the generative LLM method receive tuples, composed of an answer sentence and a set of eligible source spans $\mathcal{W}(P)$ from the paper P , annotated as factual basis for that sentence during question answering. The set of source spans is composed of floating windows of contiguous sentences in the source document, up to a length of five, and can be denoted as:

$$\mathcal{W}(P) = \bigcup_{k=1}^5 \{s_i \oplus s_{i+1} \oplus \dots \oplus s_{i+k-1} \mid 1 \leq i \leq n - k + 1\},$$

where k denotes the window size and $\oplus$ denotes string concatenation with whitespace.

Extraction Method	Retrieval Category	Candidate Selection (Top 10)	Final Selection (Top 1)
1	Native	-	SQuAI
2	Dense	-	Bi-encoder
3	Sparse	-	BM25
4	Hybrid	Bi-encoder	BM25
5	Hybrid	BM25	Bi-encoder
6	Hybrid	-	BM25 & Bi-encoder ³
7	Cross-encoder	Bi-encoder	Cross-encoder
8	Cross-encoder	BM25	Cross-encoder
9	Cross-encoder	BM25 & Bi-encoder ³	Cross-encoder
10	Generative	-	LLM

Table 3.1.: Cited text span extraction methods overview

³using Reciprocal Rank Fusion

3.2.1. Native SQuAI extraction baseline

The first method relies on SQuAI’s native text span extraction mechanism, utilizing simple lexical matching. Since it is only capable of providing one span per paper, even if it is cited repeatedly in different contexts, it is limited by design. Nevertheless, representing the attribution behaviour of the SQuAI system, it serves as a native baseline for further extraction pipelines.

3.2.2. Dense retrieval

Applying a bi-encoder, this method selects the best fitting source span out of all floating windows based on the similarity of the span’s dense vector representation with that of the answer sentence it is attributed to. It measures the suitability of relying exclusively on semantic similarity for source span attribution. As a dense retrieval baseline, it is capable of identifying appropriate source spans with different wording, but prone to overlooking precise terminology.

3.2.3. Sparse retrieval

The third method utilizes BM25, reranking source spans based on their lexical similarity with the corresponding answer sentence. The highest scoring span is then selected as final attributed span. Representing the sparse retrieval baseline BM25 is especially effective with precise terminology, however less robust to paraphrasing.

3.2.4. Hybrid retrieval

Hybrid methods aim to combine sparse and dense retrieval, leveraging their complementary strengths and weaknesses. Thus, these variants evaluate whether the retrieval performance can be enhanced by combining both approaches into a single method. The fourth method applies a two-stage retrieval method, with the bi-encoder selecting the top ten candidate windows and BM25 selecting the final span, with method five reversing their order. While the former prioritizes semantic recall during candidate selection and lexical precision during final span selection, the latter reverses this order. Comparing both allows for analysis, if either of those paradigms is better suited for candidate or final span selection. The sixth method combines both lexical and semantic matching into a single score using Reciprocal Rank Fusion (RRF). Instead of sequential application, this variant evaluates hybrid retrieval performance without an imposed order between the components themselves.

3.2.5. Cross-Encoder reranking

Overcoming the limitation of both BM25 and bi-encoder retrieval of only being able to process the answer sentence and source span in isolation, the following methods introduce cross-encoders, allowing for joint processing of both. However, as cross-encoders are far more computationally expensive than previous methods and thus not feasible to run on the whole set of floating window candidate spans, these methods instead use different candidate selectors, with the cross-encoder

selecting the best fitting final span for the answer sentence. Methods seven and eight apply semantic and lexical matching, while method nine utilizes both joined together as candidate selector.

3.2.6. Generative LLM

The last method uses a generative LLM to extract the best supporting span from the source document. Unlike previous methods, it does not rerank candidate spans, but instead prompts the model to identify the best suited text span. Representing a generative approach, it is used to test whether a LLM's reasoning capabilities can improve evidence selection.

3.3. Span evaluation

4. Evaluation

4.1. Dataset

4.2. Cited text span extraction

4.3. Span evaluation

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A. Appendix

```

1  You are generating a scientific 2-hop question-answer-evidence (Q-A) Tuple.
2  You are given 5 candidate scientific papers in the following format:
3  [
4      {
5          "ArXiv": string,
6          "text": string,
7      }
8  ]
9
10 Your task is to select the best pair of papers and generate ONE question that
    requires connected reasoning across exactly TWO papers.
11
12 PAPER SELECTION
13 First, select the best pair of papers.
14 The selected pair must satisfy:
15 -Paper A provides an intermediate entity, method, dataset, variable, or result
16 -Paper B uses, evaluates, extends, contrasts with, explains, or depends on that
    intermediate element
17 -The final answer requires combining both papers
18 -Do NOT select papers that are only loosely related or redundant.
19
20 REQUIREMENTS
21 A valid question MUST:
22 -require exactly two reasoning steps (2-hop)
23 -use exactly two supporting papers
24 -require sequential reasoning:
25     -Step 2 must depend on the result of Step 1
26     -If Step 1 is removed, Step 2 should not be solvable
27 -NOT be answerable from either paper alone
28 -be self-contained and unambiguous
29 -require combining information across both papers
30 -only use information that is present in the papers, do not invent hypothetical
    connections or applications. Only use existing connections.
31 -question cannot contain explicit references to the papers or its content such as "in
    this paper", "the proposed methods" or similar
32
33 AVOID
34 Do NOT generate:
35 -multi-part questions (e.g., "What is X and what is Y?")
36 -vague or generic questions (e.g., "implications", "advantages")
37 -literature summary questions
38 -questions answerable from a single paper
39 -questions where papers are only topically related but not logically connected
40 -Do NOT use external scientific knowledge, commonsense assumptions, or unstated
    domain knowledge, only use the information that is present in the papers

```

```

41
42 OUTPUT FORMAT
43 {
44     "usedPapers" : [
45         {
46             "arXiv": <paperId>,
47             "role": <bridgeEvidence / bridgeAnswer depending on the papers role>
48         },
49         {
50             "arXiv": <paperId>,
51             "role": <bridgeEvidence/ bridgeAnswer depending on the papers role>
52         }
53     ],
54     "rejectedPapers": [<paperId1>, <paperId2>, <paperId3>],
55     "reasoning": {
56         "step1": <Explain your reasoning for step1 on paper A>,
57         "step2": <use step1 reasoning to derive or support the final answer>,
58         "connectionExplanation": <explain why the steps are connected and why step2
59             requires the result of step1>,
60     },
61     "questionDraft" : "<one clear, self-contained question requiring 2-hop reasoning,
62         this serves as a draft for you final question. You can but don't have to
63         explicitly reference the papers here to help formulate a better question.>",
64     "question" : <decontextualize the questionDraft, so that it contains no explicit
65         reference to the papers or external figures. Every reference to a paper
66         should instead directly name the concept, phenomenon, method or similar that
67         you are referring to, briefly describing it if needed. The question cannot
68         contain explicit references to the papers or its content such as "in this
69         paper", "the proposed methods" or similar.>,
70     "answerWithPaperReferences" : <long-form paragraph integrating Paper A and Paper
71         B with citations like [Paper A], [Paper B]>,
72     "answerWithoutPaperReferences": <long-form paragraph that answers the question,
73         without referencing the papers, but directly incorporatestheir information,
74         so that it is standalone understandable>,
75     "isNotSingleHop": <explain why the question you generated is not single hop>
76 }
77
78 Do not deviate from this schema. Do not add any preciding information like ``json.
79 Only Answer with the valid json
80 Paper Texts:

```

Listing A.1: Question–Answer generation prompt

- 1 You are evaluating a scientific question-answer (Q-A) example.
- 2 Your goal is to assess whether it is a valid 2-hop, evidence-grounded scientific multi-hop question.
- 3 A valid example must:
- 4 -require combining information from exactly TWO papers

```

5  -involve dependent reasoning (Step 2 must require Step 1)
6  -not be decomposable into independent sub-questions
7  -be fully answerable from the provided evidence
8  -be self-contained and unambiguous
9
10 IMPORTANT:
11 -For each criterion, assign one label:
12 -GOOD = fully satisfies the criterion
13 -BORDERLINE = partially satisfies the criterion
14 -BAD = does not satisfy the criterion
15 -Be strict: only assign GOOD if the criterion is clearly and fully satisfied.
16
17 INPUT:
18 Question: {question}
19 Answer: {answer}
20 bridgeEvidencePaperText: {paper1Text}
21 bridgeAnswerPaperText: {paper2Text}
22 Reasoning Steps: {reasoningSteps}
23
24 EVALUATION CRITERIA
25 A. Reasoning Structure
26 Multi-hop Validity: Does answering the question require combining both papers?
27 -GOOD: Both papers are strictly required
28 -BORDERLINE: Both papers contribute but one may be sufficient
29 -BAD: Only one paper is sufficient (single-hop)
30
31 Dependency Strength: Does Step 2 depend on the result of Step 1?
32 GOOD: Step 2 strictly requires Step 1
33 BORDERLINE: Partial dependence
34 BAD: Steps are independent (disconnected reasoning)
35
36 Non-Decomposability: Is the question NOT decomposable into independent sub-questions?
37 -GOOD: Cannot be split; requires joint reasoning
38 -BORDERLINE: Partially decomposable
39 -BAD: Clearly decomposable into independent sub-questions
40
41 B. Evidence Grounding: Evidence DistributionAre both papers required and non-
    redundant?
42 -GOOD: Each paper contributes distinct, necessary information
43 -BORDERLINE: Some overlap or redundancy
44 -BAD: One paper is sufficient; the other is redundant
45
46 Answerability: Is the answer fully supported by the provided evidence?
47 -GOOD: Fully supported by cited evidence
48 -BORDERLINE: Partially supported
49 -BAD: Not supported or contradicts the evidence
50
51 C. Dataset Quality:

```

```

52 Decontextualization
53 Is the question self-contained and unambiguous? The question cannot contain explicit
    references to the papers or its content such as "in this paper", "the proposed
    methods", " this approach" or similar.
54 -GOOD: Fully self-contained; all entities clearly defined
55 -BORDERLINE: Minor ambiguity
56 -BAD: Not understandable without external context
57
58 OUTPUT FORMAT
59 {{
60     "multiHopValidity": {{
61         "judgement": "<GOOD / BORDERLINE / BAD>",
62         "explanation": "<explain your judgement for multiHopValidity>"
63     }},
64     "dependencyStrength": {{
65         "judgement": "<GOOD / BORDERLINE / BAD>",
66         "explanation": "<explain your judgement for dependencyStrength>"
67     }},
68     "nonDecomposability": {{
69         "judgement": "<GOOD / BORDERLINE / BAD>",
70         "explanation": "<explain your judgement for nonDecomposability>"
71     }},
72     "evidenceDistribution": {{
73         "judgement": "<GOOD / BORDERLINE / BAD>",
74         "explanation": "<explain your judgement for evidenceDistribution>"
75     }},
76     "answerability": {{
77         "judgement": "<GOOD / BORDERLINE / BAD>",
78         "explanation": "<explain your judgement for answerability>"
79     }},
80     "decontextualization": {{
81         "judgement": "<GOOD / BORDERLINE / BAD>",
82         "explanation": "<explain your judgement for decontextualization>"
83     }},
84     "confidence": <0 to 1 rate how confident you are in the judgements>
85 }}
86
87 Do not deviate from this schema. Do not add any preciding information like ‘‘‘json.
    Only Answer with the valid json

```

Listing A.2: Judgement prompt

```

1 You are given 2 scientific paper texts as well as a logical connection that exists
    between that papers. Your task is to determine wether that connection is
    explicitelty mentioned in one of the 2 papers, or if is drawn based on their
    content, but not explicitelty mentioned.

```

```
2
3  IMPORTANT:
4  -Use only the information from the given papers
5  -Do NOT rely on external knowledge
6  -Be strict: if any critical information is missing, the answer is NOT recoverable
7  -Partial or speculative answers count as NOT answerable
8
9  INPUT:
10 paper1: {paper1}
11 paper2: {paper2}
12 connection: {connection}
13
14 OUTPUT FORMAT
15 {{
16     "explanation" : "<describe your reasoning>",
17     "isPresent": <true/ false depending on if the connection is explicitly mentioned
18         in at least one of the papers, false if not>
19 }}
20 Do not deviate from this schema. Do not add any preciding information like ‘‘‘json.
21     Only Answer with the valid json
```

Listing A.3: DIRE prompt